

Assignment 2: ECO764

Abhishek L
Roll No. 230049

Ayush Gupta
Roll No. 230257

Neel D Jadia
Roll No. 230688

Priyanka Arora
Roll No. 230799

Shreyas Chougaonkar
Roll No. 230990

1: Idiosyncratic Volatility and Stock Returns

a: Data and Descriptive Statistics:

The study selects 10 prominent large-cap stocks from the National Stock Exchange (NSE) of India, including RELIANCE, TCS, and HDFCBANK, with the Nifty 50 index acting as the market proxy. Spanning from January 2014 to March 2026, the daily adjusted closing prices provide a comprehensive look at market behavior through log returns, computed as $r_{i,t} = \ln(P_{i,t}/P_{i,t-1})$. Statistical analysis (Table 1) reveals that all selected series exhibit excess kurtosis significantly higher than 3, which is a clear indicator of fat-tailed distributions. This empirical evidence suggests that these stocks are prone to extreme price movements and asymmetric downside risk, aligning with the typical stylized facts of emerging equity markets.

Table 1: Descriptive Statistics of Daily Log Returns (2014–2026)

Stock	Mean	Std Dev	Skewness	Kurtosis
RELIANCE	0.000677	0.016882	0.080051	11.042367
TCS	0.000361	0.014887	-0.179598	6.929687
HDFCBANK	0.000508	0.013906	-0.283718	13.190049
INFY	0.000451	0.016720	-0.591665	12.493012
ICICIBANK	0.000617	0.018938	-0.073806	12.130278
HINDUNILVR	0.000511	0.013913	0.566070	9.722597
SBIN	0.000584	0.020339	0.438601	14.812040
BAJFINANCE	0.001305	0.022348	-0.559401	13.287394
WIPRO	0.000263	0.016118	0.193717	9.277170
AXISBANK	0.000449	0.020396	-1.239953	30.277489
NIFTY50	0.000429	0.010146	-1.304250	22.761797

b: CAPM Estimation and Idiosyncratic Volatility:

We estimate the Capital Asset Pricing Model (CAPM) using Ordinary Least Squares (OLS) regression: $R_{i,t} - R_{f,t} = \alpha_i + \beta_i(R_{m,t} - R_{f,t}) + \epsilon_{i,t}$. The idiosyncratic volatility ($IVOL_i$) is calculated as the standard deviation of the OLS residuals, representing firm-specific risk that remains after accounting for market movements. Our results (Table 2) show betas ranging from 0.56 for HINDUNILVR to 1.36 for AXISBANK, illustrating the distinct defensive and cyclical characteristics of the chosen stocks. Specifically, BAJFINANCE displays the highest $IVOL$ at 0.017967, whereas HDFCBANK maintains the lowest at 0.009519, indicating a wide variance in firm-specific risk across the portfolio.

Table 2: CAPM Estimates and Idiosyncratic Volatility (2014–2026)

Stock	$\hat{\alpha}$	$\hat{\beta}$	IVOL
RELIANCE	0.000230	1.095837	0.012704
TCS	-0.000004	0.662536	0.013283
HDFCBANK	0.000079	0.999145	0.009519
INFY	0.000061	0.795992	0.014640
ICICIBANK	0.000124	1.335398	0.013232
HINDUNILVR	0.000166	0.562043	0.012691
SBIN	0.000093	1.325710	0.015257
BAJFINANCE	0.000817	1.309835	0.017967
WIPRO	-0.000111	0.709627	0.014420
AXISBANK	-0.000047	1.354873	0.015068

c: Portfolio Sorting by Idiosyncratic Volatility

Stocks are sorted into three equally sized portfolios: Low, Medium, and High IVOL, using the 33rd and 67th percentiles of the IVOL distribution as breakpoints. Equal-weighted average daily returns are computed for each portfolio. As shown in Table 3, the High IVOL portfolio earns the highest average return (0.000779), followed by Low IVOL (0.000578) and Medium IVOL (0.000358), suggesting a positive relationship between idiosyncratic risk and returns.

Table 3: IVOL Portfolio Composition and Average Daily Returns

Portfolio	Stocks	Avg. Daily Return
Low IVOL	RELIANCE, HDFCBANK, ICICIBANK, HINDUNILVR	0.000578
Medium IVOL	TCS, INFY, WIPRO	0.000358
High IVOL	SBIN, BAJFINANCE, AXISBANK	0.000779

d: Cross-Sectional Regression and the Idiosyncratic Volatility Puzzle

To formally test the IVOL–return relationship, we estimate the following cross-sectional regression where R_i is the time-series average daily return of stock i :

$$R_i = \gamma_0 + \gamma_1 IVOL_i + u_i$$

A positive and significant $\hat{\gamma}_1$ supports a conventional risk-return tradeoff, while a negative $\hat{\gamma}_1$ would support the idiosyncratic volatility puzzle. Table 4 presents the results.

Table 4: Cross-Sectional Regression: Returns on Idiosyncratic Volatility

Variable	Estimate	Std. Error	t-value	p-value
Intercept ($\hat{\gamma}_0$)	-0.0003643	0.0005481	-0.665	0.525
IVOL ($\hat{\gamma}_1$)	0.0674963	0.0390544	1.728	0.122
$R^2 = 0.2719$		Adjusted $R^2 = 0.1808$		
$F = 2.987$		$p\text{-value} = 0.1222$		

The estimated coefficient $\hat{\gamma}_1 = 0.0675$ is positive, indicating that stocks with higher idiosyncratic volatility tend to earn higher average returns. However, the coefficient is statistically insignificant ($p = 0.122$), and the model explains only 27% of cross-sectional variation in returns across the 10 stocks. These results **do not support the idiosyncratic volatility puzzle**; instead, they are consistent with a conventional risk-return tradeoff, though the small cross-section ($n = 10$) and lack of statistical significance preclude strong conclusions.

2: Empirical Analysis of Factor Investing

a: Data Description and Return Construction

The empirical analysis utilizes monthly closing prices for 20 NSE-listed stocks and the Nifty 50 Index for the period from January 2013 to December 2024, sourced primarily from Yahoo Finance. Monthly returns are computed using the standard percentage change formula to capture the capital gains over time. This dataset allows for a robust examination of how different firm characteristics drive returns in the Indian equity landscape. By standardizing these returns, we ensure that our subsequent factor portfolio construction is grounded in a consistent and comparable historical framework.

b: Factor Portfolio Construction (MOM & LOWVOL)

Summary Statistics:

Two distinct factor portfolios, Momentum (MOM) and Low Volatility (LOWVOL), are constructed by ranking stocks based on their past 12-month performance and rolling standard deviations. Stocks in the top 30% are used to form the "Winner" and "High Volatility" groups, while the bottom 30% form the "Loser" and "Low Volatility" groups. The factors are then calculated as the difference between these extreme portfolios to isolate the specific premia associated with these anomalies. This methodological framework follows the established literature on factor investing, aiming to test if persistence or defensive traits lead to superior risk-adjusted performance.

Table 5: Summary Statistics of Monthly Returns (All Stocks)

Stock	Count	Mean	Std Dev	Min	Max	Skew / Kurt
ADANIENT	143	0.0459	0.1752	-0.5427	0.7326	0.51 / 3.55
ASIANPAINT	143	0.0148	0.0745	-0.1707	0.2479	0.46 / 0.61
AXISBANK	143	0.0140	0.0949	-0.4565	0.2357	-0.59 / 3.61
BAJFINANCE	143	0.0351	0.1141	-0.5027	0.4835	0.19 / 5.32
HDFCBANK	143	0.0146	0.0612	-0.2681	0.2174	-0.23 / 2.98
HINDUNILVR	143	0.0145	0.0626	-0.1454	0.2502	0.79 / 1.51
ICICIBANK	143	0.0171	0.0846	-0.3489	0.2685	-0.21 / 2.24
INFY	143	0.0168	0.0739	-0.2263	0.3126	0.03 / 1.85
ITC	143	0.0103	0.0606	-0.1723	0.1719	-0.03 / 0.55
KOTAKBANK	143	0.0138	0.0674	-0.2001	0.2325	0.17 / 0.88
LT	143	0.0161	0.0800	-0.3097	0.2345	-0.08 / 1.38
MARUTI	143	0.0180	0.0874	-0.3175	0.3032	-0.16 / 1.89
NTPC	143	0.0128	0.0808	-0.2048	0.3795	0.94 / 2.86
POWERGRID	143	0.0164	0.0588	-0.1156	0.1618	0.20 / -0.39
RELIANCE	143	0.0160	0.0778	-0.1617	0.3163	0.74 / 1.20
SBIN	143	0.0141	0.1009	-0.3503	0.3830	0.31 / 1.75
TCS	143	0.0167	0.0664	-0.1252	0.2397	0.59 / 0.32
TITAN	143	0.0217	0.0894	-0.2557	0.2891	-0.26 / 0.73
ULTRACEMCO	143	0.0158	0.0777	-0.2310	0.2268	0.15 / 0.47
WIPRO	143	0.0131	0.0760	-0.2031	0.2791	0.43 / 1.43

c: Multi-Factor Asset Pricing Model

We estimate a multi-factor regression model for RELIANCE to determine its exposure to market, momentum, and low-volatility factors. The OLS results indicate that the market factor is highly significant ($p < 0.01$), confirming the stock's strong sensitivity to overall economic shifts. Additionally, the momentum factor is statistically significant at the 5% level, suggesting that RELIANCE tends to follow established return trends. In contrast, the low-volatility factor remains insignificant, implying that the stock does not carry a defensive premium. The model explains 44.2% of the return variation, highlighting the effectiveness of these factors in describing security-level performance.

Table 6: OLS Regression Results for RELIANCE

Variable	Coefficient	Std Error	t-Statistic	p-value
Intercept (α)	-0.0094	0.0070	-1.311	0.192
Market (MKT)	1.2005	0.1610	7.474	0.000
Momentum (MOM)	0.3493	0.1170	2.990	0.003
Low Volatility	0.0764	0.1240	0.615	0.540

d: Performance Evaluation

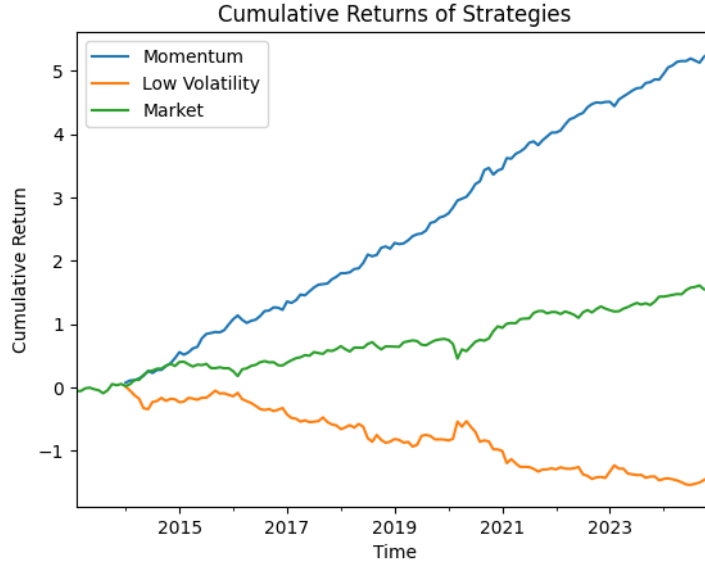


Figure 1: Cumulative Returns of Momentum, Low Volatility, and Market Portfolio

The comparative analysis of factor portfolios reveals that the momentum strategy (Sharpe: 0.8472) significantly outperforms the broad market benchmark. This indicates that price persistence is a strong driver of alpha in the NSE over the studied decade. Conversely, the low-volatility strategy underperforms with a negative Sharpe ratio of -0.1822, suggesting that the "low-volatility anomaly" does not hold in this context. These findings emphasize that higher-risk, high-momentum stocks have historically delivered better returns in the Indian market than defensive, low-risk counterparts.

3: Construction of Continuous Futures Series

a: Multi-Maturity Data Preparation

For the construction of a reliable futures series, historical prices data covering April 2015 to March 2025 was meticulously aggregated from Investing.com and MCX India to ensure accuracy across various maturities. This multi-maturity approach allows us to observe the term structure of futures prices, providing insights into the cost-of-carry and market expectations of future gold prices. By selecting these overlapping contracts, we lay the necessary foundation for executing a seamless rollover process, which is critical for long-term econometric analysis.

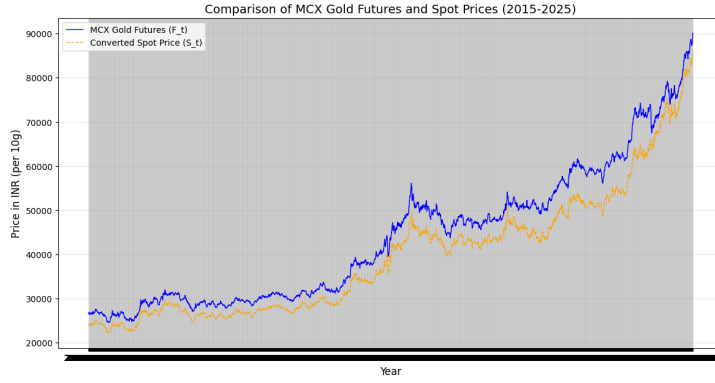


Figure 2: Synchronization of Gold Spot and MCX Futures (2015–2025)

b: Continuous Series Construction (Rollover Method)

A continuous futures price series was constructed using a calendar-based rollover methodology to avoid the artificial price jumps associated with contract expiration. Specifically, the position is transitioned from the Near-month contract to the Next-month contract exactly five trading days prior to the expiration of the front-month contract. This method ensures that the resulting price series reflects the actual returns available to a long-term investor who rolls their positions forward. By smoothing these transitions, the continuous series becomes a robust proxy for the futures market, suitable for high-precision unit root and cointegration testing.

c: Unit Root Testing (ADF Test)

The Augmented Dickey-Fuller (ADF) test was applied to both the Spot and Continuous Futures price series to determine their order of integration. In both cases, the test results for price levels failed to reject the null hypothesis of a unit root ($p \approx 1.00$), indicating non-stationarity. However, upon first differencing, the series became stationary ($p < 0.01$), proving that both prices are integrated of order one, or $I(1)$. This statistical property is a prerequisite for cointegration, as it confirms that the series share a stochastic trend that could potentially be tied together in a long-run equilibrium.

Table 7: ADF Unit Root Test Results

Variable	ADF Statistic	p-value	Conclusion
Futures Price (Levels)	2.6654	0.9991	Non-Stationary
Spot Price (Levels)	3.1513	1.0000	Non-Stationary
Futures (1st Diff)	-14.2669	0.0000	Stationary
Spot (1st Diff)	-20.2587	0.0000	Stationary

d: Cointegration Analysis

The Johansen Trace Statistic of 18.3217 exceeds the 95% critical value of 15.4943, providing strong empirical evidence that a long-run equilibrium exists between Gold Spot and Futures prices. This result signifies that the two markets are fundamentally "tied" together, meaning they do not drift infinitely far apart over time despite short-term fluctuations. From an economic perspective, this cointegration is driven by arbitrageurs who exploit price differences, forcing the two series back toward a common stochastic trend. This ensures that the Indian gold market remains efficient, with the derivative and physical markets correctly reflecting the same underlying asset value in the long term.

4: VECM and Market Dynamics

a: Vector Error Correction Model Estimation

The estimation of the Vector Error Correction Model (VECM) reveals a statistically significant and stable long-run relationship where $Spot = 1.1563 \times Futures$. The loading coefficients (α) for both markets are negative and highly significant, confirming that both Futures (-0.0137) and Spot (-0.0119) actively adjust to correct deviations from the equilibrium.

Table 8: VECM Long-Run Relationship and Speed of Adjustment

Metric	Coefficient	Economic Interpretation
Cointegrating Vector (β)	1.1563	Stable Long-Run Price Anchor
Futures Adj. Speed (α_f)	-0.0137	1.37% Daily Correction to Equilibrium
Spot Adj. Speed (α_s)	-0.0119	1.19% Daily Correction to Equilibrium
Statistical Significance	$p < 0.01$	Strong evidence of active mean reversion

This adjustment process ensures that any short-term "basis" divergence is temporary and eventually mean-reverting. The significant t-statistics indicate that both markets are responsive to the long-run price trend, although the slightly higher adjustment speed in the futures market suggests it may react more quickly to correct pricing inefficiencies.

b: Price Discovery Analysis

Our leadership evaluation, using Component Share (CS), Information Share (IS), and Garbade-Silber (GS) measures, provides an internally consistent view of market dominance. The results indicate that the Spot market is the primary driver of price discovery in India, accounting for 53.47% of the permanent price trend.

Table 9: Price Discovery Share and Leadership Measures

Measure	Spot Market	Futures Market	Leader
Component Share (CS)	53.47%	46.53%	Spot
Hasbrouck Info Share (IS)	High (Primary Source)	Low (Recipient)	Spot
Garbade-Silber (GS)	Central (Leader)	Satellite (Follower)	Spot

This finding is unique compared to international markets, where futures typically lead, and it likely reflects the heavy influence of local physical demand and regional import policies. Because the spot price responds directly to physical supply shocks and government regulations first, it acts as the "Central" market, while the futures market follows as a "Satellite" receiver of this information.

c: Optimal Hedging Strategy

To determine the most effective risk management approach, we calculated the Minimum Variance Hedge Ratio (h^*) of 0.9027, with a corresponding hedge effectiveness (R^2) of 0.6784.

Table 10: Minimum Variance Hedge Strategy Summary

Strategy Component	Estimated Value / Specification
Optimal Hedge Ratio (h^*)	0.9027
Hedge Effectiveness (R^2)	0.6784 (68% Variance Reduced)
Recommended Position	Short MCX Gold Futures
Exposure Coverage	90.27% of Physical Spot Holdings

This ratio implies that for every unit of physical gold held, an investor should take a short position of approximately 90.27% in MCX Gold Futures to minimize price risk. The high R^2 indicates that this strategy is capable of eliminating nearly 68% of the variance in spot returns, demonstrating that the futures market is a highly efficient tool for Indian gold hedgers.

5: Empirical Identification of Arbitrage Opportunities

a: Theoretical futures price estimation

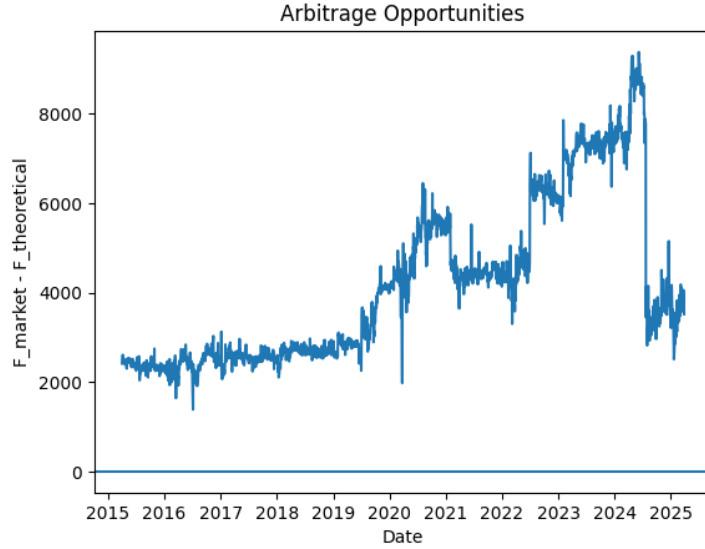
To determine the theoretical future price, the rate of interest has been proxied by short-term Treasury bill yields obtained from Yahoo Finance. Since storage cost is not given, a constant value of $c=0.02$ has been assumed. The csv file (analyzed_data) in the zip contains all the theoretical values.

b and c: Arbitrage Opportunities

The observed futures prices are compared with theoretical values derived from the cost-of-carry model. The deviation is defined as:

$$\text{Arbitrage}_t = F_t^{\text{market}} - F_t^{\text{theoretical}}$$

When $F_t^{\text{market}} > F_t^{\text{theoretical}}$, we have cash-and-carry arbitrage (i.e., $\text{Arbitrage}_t > 0$), and when $F_t^{\text{market}} < F_t^{\text{theoretical}}$, we have reverse cash-and-carry arbitrage. Arbitrage has been plotted below for each time period, and it has been found that cash-and-carry arbitrage always exists, whereas there is no period of reverse cash-and-carry arbitrage. This might also be due to the non-consideration of some other factors like transaction costs, due to which the theoretical value is coming out low.



d: Butterfly Effect

Taking the data from MCxIndia.com, on 27th March 2025 we see pricing efficiency accross multiple contracts of gold futures (with expiry dates 02 April 2026, 05 June 2026, and 05 August 2026) using butterfly spread. The butterfly spread is defined as:

$$\text{Butterfly}_t = F_{1,t} - 2F_{2,t} + F_{3,t}$$

where $F_{1,t}$, $F_{2,t}$, and $F_{3,t}$ represent the futures prices of near, mid, and far maturity contracts, respectively. The corresponding closing prices are:

$$F_{1,t} = 144282, \quad F_{2,t} = 147255, \quad F_{3,t} = 150132$$

$$\text{Butterfly}_t = 144282 - 2(147255) + 150132 = -96$$

The butterfly spread is negative. Therefore, a trader can take a long position in the near and far maturity contracts and a short position in twice the mid-maturity contract. As the prices adjust towards equilibrium, the trader is expected to earn a profit of 96 rupees from the relative mispricing across maturities.

6: Climate and Freight Derivatives Analysis

a: Temperature Dataset Analysis (Kanpur, U.P.)

The empirical analysis utilizes a historical time series of daily average temperatures for Kanpur, Uttar Pradesh, spanning a 10-year horizon from January 2016 to December 2025, sourced from the NASA POWER database. Kanpur experiences severe thermal variance, making it an ideal candidate for cooling-based temperature derivatives.

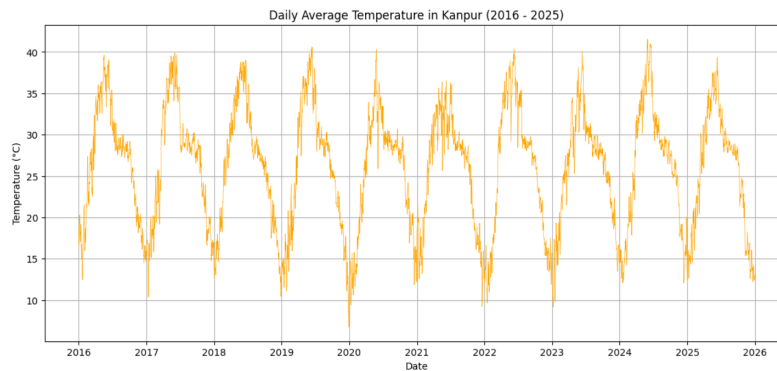


Figure 3: Daily Average Temperature in Kanpur (2016–2025)

I. Summary Statistics:

The daily average temperature exhibits a mean of 25.96°C with a standard deviation of 7.28°C , reflecting extreme seasonal variations. The negative skewness (-0.22) and platykurtic distribution (-0.86) indicate extreme cold days are marginally more dispersed and temperatures cycle widely across the decadal range.

Table 11: Summary Statistics of Daily Average Temperature (2016–2025)

Statistic	Value	Statistic	Value
Mean	25.96°C	Minimum	6.77°C
Std Dev	7.28°C	Maximum	41.55°C
Skewness	-0.22	Kurtosis	-0.86

II. Seasonality and Variance Analysis:

Boxplot analysis confirms strong deterministic seasonality. Transitional months exhibit high year-over-year unpredictability, while peak summer months (May/June) show tight

clustering of extreme high temperatures above 35°C. Long-term trends observed in decadal heatmaps visually confirm the persistent upward pressure on cooling demand, validating the necessity for temperature hedging instruments in the region.

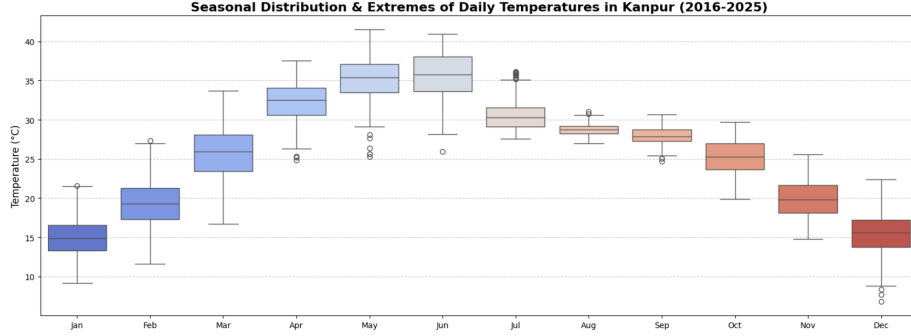


Figure 4: Seasonal Distribution of Kanpur Temperatures

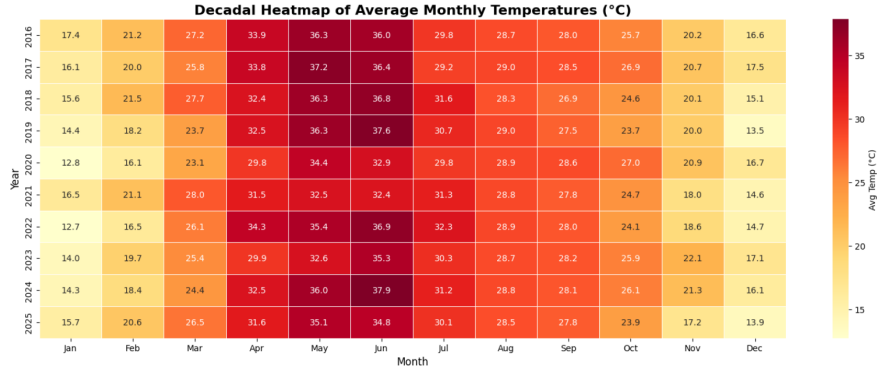


Figure 5: Decadal Monthly Temperature Heatmap

b: Model Estimation and Forecasting

I. Stationarity Analysis:

The Augmented Dickey-Fuller (ADF) test yielded a statistic of -4.6729 ($p = 0.00095$), rejecting the null hypothesis of a unit root. This confirms the daily temperature series is mean-reverting $I(0)$, though profound seasonality necessitates a Seasonal ARIMA (SARIMA) approach to neutralize annual variance.

II. Parameter Identification:

The ACF exhibits slow persistent decay, while the PACF demonstrates a highly significant positive spike at Lag 1 and 2 before collapsing, dictating an Auto-Regressive structure.

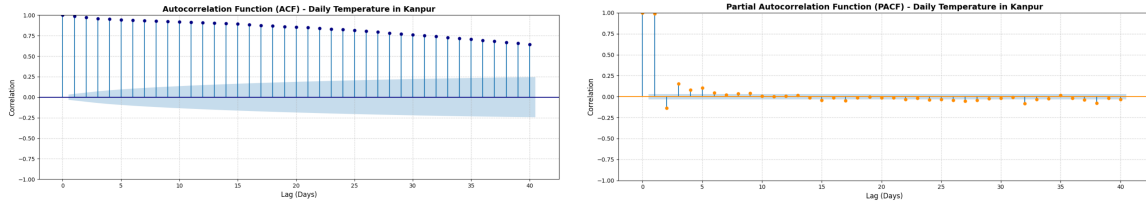


Figure 6: ACF (left) and PACF (right) Analysis for Parameter Identification

III. SARIMA Estimation and Diagnostics:

A SARIMA(1, 0, 1)(1, 1, 1)₁₂ model was estimated on monthly aggregated data. The AR.L1 coefficient (0.538) is highly significant ($p < 0.01$). Residual diagnostics and the Ljung-Box test ($p = 0.85$) confirm that residuals resemble pure white noise, indicating a well-specified model.

```

Estimating SARIMA(1,0,1)(1,1,1)[12] Model...
                                SARIMAX Results
=====
Dep. Variable:                  T2M      No. Observations:      120
Model:                        SARIMAX(1, 0, 1)x(1, 1, 1, 12)  Log Likelihood        -157.939
Date:                          Tue, 31 Mar 2026              AIC                325.877
Time:                          16:54:24                     BIC                338.594
Sample:                        01-31-2016                    HQIC               331.014
                                - 12-31-2025
Covariance Type:                opg
=====
              coef      std err          z      P>|z|      [0.025      0.975]
-----
ar.L1          0.5380        0.114        4.719      0.000        0.315        0.761
ma.L1          0.1343        0.149         0.902      0.367       -0.158        0.426
ar.S.L12       -0.0550        0.133       -0.415      0.678       -0.315        0.205
ma.S.L12       -1.0000       1480.101       -0.001      0.999     -2901.944     2899.944
sigma2          1.2931       1914.035         0.001      0.999     -3750.146     3752.732
=====
Ljung-Box (L1) (Q):              0.03   Jarque-Bera (JB):              0.76
Prob(Q):                        0.85   Prob(JB):                0.69
Heteroskedasticity (H):          0.54   Skew:                   0.01
Prob(H) (two-sided):            0.09   Kurtosis:               2.56
=====

```

Figure 7: SARIMA Residual Diagnostics

IV. Out-of-Sample Forecasting:

A 24-month forecast (2026–2027) demonstrates a highly stable mean prediction path replicating anticipated seasonal peaks and troughs. The narrow 95% confidence interval confirms strong predictive reliability for pricing climate-indexed derivatives.

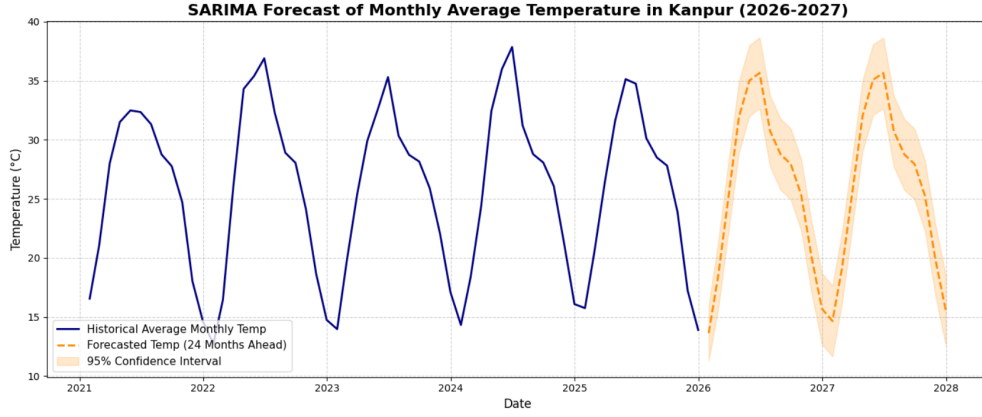


Figure 8: SARIMA Forecast of Monthly Average Temperature (2026–2027)

c: Construction of the Derivative Index

I. Index Selection and Baseline:

Due to high volumetric risk during summer heatwaves, we construct a **Cooling Degree Day (CDD)** index using the standard baseline temperature $T_{\text{base}} = 18^\circ\text{C}$, capturing days with significant cooling demand.

II. Mathematical Aggregation:

The Daily CDD is computed using the non-linear payoff function:

$$CDD_t = \max(0, T_t - 18)$$

The cumulative index for May (the target accumulation period) is defined as:

$$Index_{May,Y} = \sum_{t=1}^{31} CDD_t$$

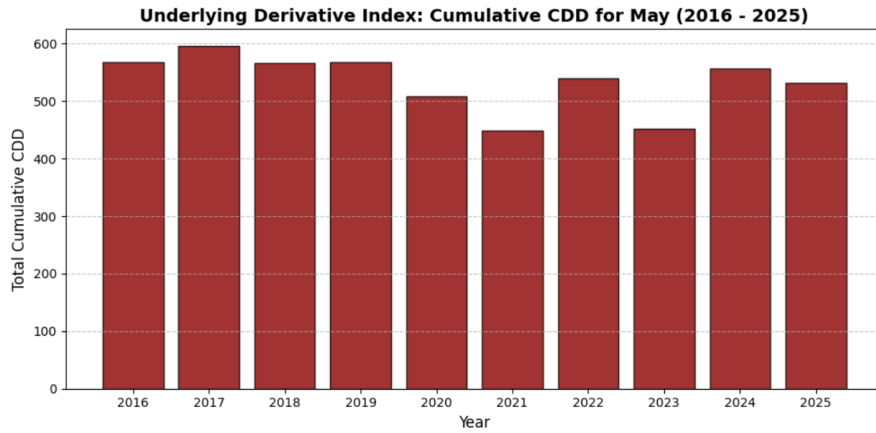


Figure 9: Cumulative CDD Index Aggregation

d: Derivative Contract Design and Payoff Simulation

I. Contract Specifications:

We design a European-style Weather Call Option for May 2026 with a Strike Level (K) of 560 CDD. The tick size is 10,000 per CDD point, with a payout cap at 10,00,000 (reached at 660 CDD).

II. Payoff Function:

$$Payoff = \min \left(10,00,000, 10,000 \times \max(0, CDD_{actual} - 560) \right)$$

III. Scenario Simulation:

Simulations demonstrate the contract's effectiveness for energy providers. In Scenario A (Heatwave), a realized index of 610 CDD triggers a 5,00,000 payout, effectively hedging increased operational costs. In Scenario B (Normal May), the option expires out-of-the-money, as intended.

Table 12: Derivative Payout Simulation for May 2026

Weather Scenario	Realized Index (CDD)	Financial Payout
Scenario A: Severe Heatwave	610	5,00,000.00
Scenario B: Normal May	535	0.00
Scenario C: Unusually Cool May	480	0.00

7: Destabilisation Effect of Futures Markets on Spot Prices

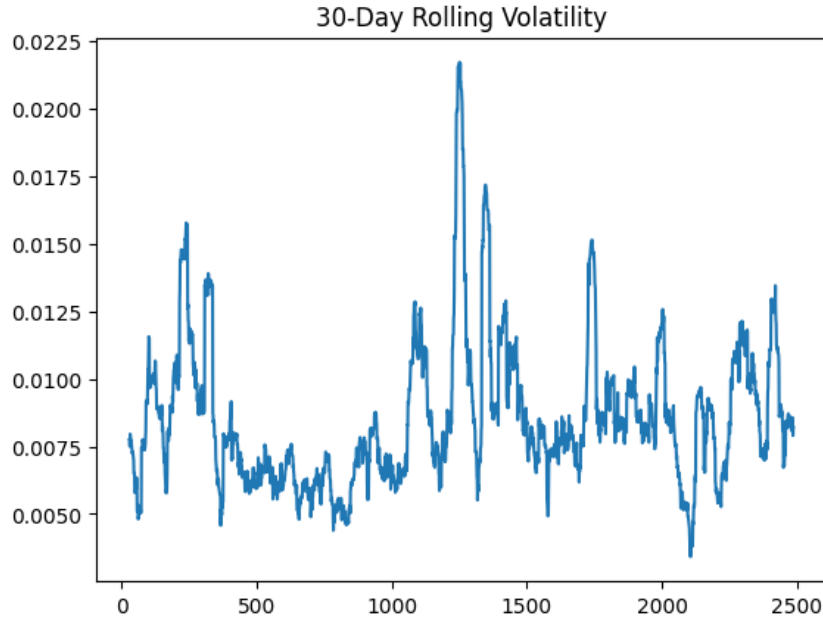
a: Daily return volatility

Spot price volatility is computed using a 30-day rolling standard deviation of log returns. This is shown in the CSV (analyzed_data) attached and the plot is given below.

b: GARCH(1,1) model for the spot price returns

The estimated GARCH(1,1) model shows that the ARCH coefficient ($\alpha = 0.0501$) is positive and significant, which means recent shocks affect current volatility. The GARCH coefficient ($\beta = 0.9299$) is also positive and very high, indicating strong persistence in volatility. The sum of the coefficients is:

$$\alpha + \beta = 0.0501 + 0.9299 = 0.98$$



Since this is close to 1, it shows that volatility is highly persistent and there is strong volatility clustering in the spot price. Volatility in spot prices is persistent, meaning shocks last for a long time.

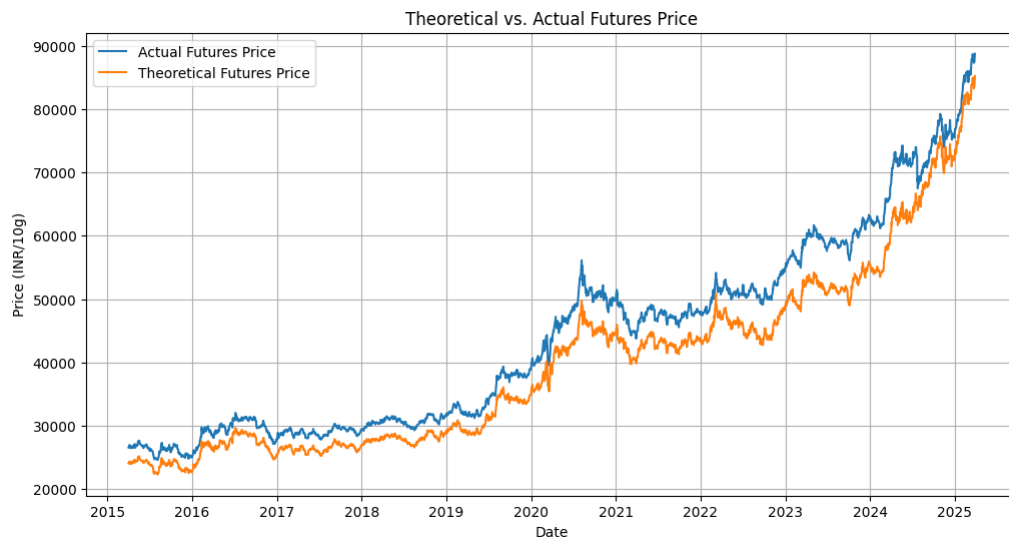
c: GARCH model including futures return

To examine the impact of futures markets on spot price volatility, the GARCH model is extended by including futures returns as an explanatory variable in the variance equation. However, based on the estimated results, the core GARCH parameters remain largely unchanged, with $\alpha = 0.0501$ and $\beta = 0.9299$, indicating strong persistence in volatility. Since the sum $\alpha + \beta = 0.98$ is close to 1, volatility clustering continues to be observed even after including futures information. The results suggest that volatility in spot prices remains highly persistent. The inclusion of futures market information does not significantly alter the volatility dynamics, indicating that futures trading does not have a strong additional effect on spot volatility in this specification.

d: Comparison of Volatility

To examine the destabilisation effect, volatility is compared across periods of high and low futures market activity. I have taken the median of future volatility as a divider and divided the data into high future volatility and low future volatility. The average volatility during periods of high futures activity is 0.008589, while during periods of low futures activity is 0.008482. Since volatility is slightly higher during periods of high futures activity, this suggests that futures trading may have a mild destabilising effect on the spot market. This result proves **destabilisation hypothesis**.

1 Appendix



Data sources:

1. Investing.com
2. Kaggle
3. Yfinanace
4. MCXINDIA